

CALVINIA, South Africa

WMO: 686180

Lat: 31.4819S

Lon: 19.7617E

Elev: 977

StdP: 90.13

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.3	0.0	-12.5	1.4	19.6	-10.7	1.7	18.8	12.3	12.4	11.1	13.0	2.5	80	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	35.7	16.1	34.2	15.8	32.9	15.4	18.7	28.3	18.0	28.3	17.2	27.9	5.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.9	21.5	14.7	11.8	21.2	13.5	10.8	20.9	57.3	27.9	54.7	28.3	52.2	27.6	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.3	8.4	-3.5	38.6	1.1	1.1	-4.3	39.4	-4.9	40.0	-5.5	40.6	-6.3	41.4
			-4.6	20.2	1.2	1.0	-5.5	20.9	-6.2	21.5	-6.9	22.0	-7.8	22.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	22.9	23.0	21.0	17.5	13.8	10.7	10.3	10.6	13.2	16.7	18.5	21.0	(6)
(7)		DBStd	5.98	3.52	3.31	3.53	4.09	3.97	3.53	3.96	4.03	4.11	4.39	4.03	3.42	(7)
(8)		HDD10.0	146	0	0	0	1	10	33	44	41	13	2	1	0	(8)
(9)		HDD18.3	1265	5	3	14	64	149	231	250	241	163	86	46	13	(9)
(10)		CDD10.0	2537	399	363	340	225	128	53	54	61	108	208	256	341	(10)
(11)		CDD18.3	614	147	133	96	38	8	1	2	2	8	34	51	95	(11)
(12)		CDH23.3	8815	1997	1729	1264	514	121	6	7	52	196	647	874	1408	(12)
(13)		CDH26.7	3857	990	842	543	166	16	0	0	8	50	256	358	628	(13)
(14)	Wind	WSAvg	4.1	4.2	4.0	3.8	3.9	3.7	4.1	4.3	4.2	4.0	4.3	4.3	4.2	(14)
(15)	Precipitation	PrecAvg	204	8	10	22	23	22	31	23	21	13	10	12	11	(15)
(16)		PrecMax	467	64	76	98	120	78	119	94	54	70	59	86	55	(16)
(17)		PrecMin	91	0	0	0	0	0	1	0	0	0	0	0	0	(17)
(18)		PrecStd	72	14	15	23	25	21	24	22	15	14	12	19	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	37.5	35.7	32.8	28.9	24.1	24.3	27.5	31.0	34.4	35.6	36.1	(19)
(20)			MCWB	16.8	16.9	16.3	15.5	13.3	10.8	10.9	11.2	12.8	14.9	14.9	15.6	(20)
(21)		2%	DB	35.6	35.2	33.5	30.5	26.5	22.3	22.3	24.6	28.2	31.8	32.6	34.0	(21)
(22)			MCWB	16.2	16.4	15.9	14.8	12.2	10.2	10.1	10.6	12.2	14.1	14.2	15.2	(22)
(23)		5%	DB	33.7	33.6	31.9	28.7	24.6	20.8	20.7	22.2	25.8	29.8	30.6	32.2	(23)
(24)			MCWB	15.9	16.2	15.1	14.0	11.8	9.6	9.3	9.9	11.3	13.3	13.7	14.9	(24)
(25)		10%	DB	32.0	31.9	30.0	26.5	22.6	18.9	18.8	19.9	23.1	27.2	28.4	30.4	(25)
(26)			MCWB	15.3	15.8	14.6	13.5	11.1	8.9	8.7	8.9	10.3	12.4	13.0	14.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	20.1	19.1	17.5	14.7	12.4	12.3	13.1	15.0	16.6	17.6	18.5	(27)
(28)			MCDWB	27.7	28.4	27.9	26.8	23.2	19.8	21.5	21.3	24.9	29.1	28.7	28.1	(28)
(29)		2%	WB	18.4	18.9	18.1	16.5	13.6	11.3	10.9	11.6	13.3	15.3	16.1	17.3	(29)
(30)			MCDWB	29.0	28.3	27.1	25.7	22.9	18.7	19.6	21.6	24.4	28.4	26.3	28.2	(30)
(31)		5%	WB	17.6	18.1	17.2	15.6	12.7	10.5	10.0	10.6	12.1	14.3	14.9	16.5	(31)
(32)			MCDWB	29.1	28.1	26.2	25.3	21.8	17.9	18.5	20.1	23.6	27.2	26.5	27.9	(32)
(33)		10%	WB	16.7	17.2	16.3	14.6	11.9	9.7	9.2	9.5	10.9	13.2	13.9	15.7	(33)
(34)			MCDWB	28.5	28.3	25.9	24.0	20.3	16.7	17.0	18.4	21.5	25.5	25.9	27.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	16.1	15.9	14.5	13.3	12.5	12.7	13.7	15.3	16.5	16.1	16.3	(35)
(36)			MCDBR	17.6	17.7	18.1	17.5	16.3	15.4	15.3	17.6	19.6	20.1	19.2	18.7	(36)
(37)			MCWBR	6.1	6.7	6.8	7.0	6.8	7.0	7.0	7.7	8.2	8.4	7.2	6.8	(37)
(38)		5% WB	MCDWB	14.6	14.2	13.9	14.6	13.4	12.6	13.2	15.6	17.1	17.3	16.8	15.8	(38)
(39)			MCWBR	5.2	5.4	5.8	6.1	6.0	6.5	6.6	7.3	7.5	7.6	6.7	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.303	0.297	0.283	0.269	0.259	0.262	0.281	0.293	0.305	0.288	0.291	(40)	
(41)		taud	2.617	2.603	2.612	2.665	2.682	2.672	2.646	2.557	2.489	2.490	2.609	2.621	(41)	
(42)		Ebn at Noon	1044	1019	992	959	927	914	924	944	979	1005	1046	1052	(42)	
(43)		Edh at Noon	102	100	93	79	69	65	70	86	103	111	102	102	(43)	
(44)	All-Sky Solar Radiation	RadAvg	8.64	7.80	6.46	4.87	3.68	3.10	3.44	4.30	5.73	7.18	8.29	8.89	(44)	
(45)		RadStd	0.24	0.24	0.20	0.24	0.21	0.18	0.15	0.20	0.25	0.27	0.31	0.35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A	(46)
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page